Retimer Market – Broadcom, Astera Labs, Marvell Technology & Nvidia – Former Sales Executive at Astera Labs Inc

Former Employee | 11 February 2025

Specialist Background

- Over 20 years' experience in the data centre space
- Can speak to Astera Labs' competitive dynamics and strengths and weaknesses
- Well-placed to discuss share dynamics for retimers and key competitors such as Broadcom, Marvell Technology and Nvidia

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Analyst:
Astera, we founded in 2017, a pretty new company, PCIe [peripheral component interconnect express] native. When you talk to a lot of investors, really great retimer product, first to market, visionary and being able to recognise that it was going to be necessary at Hopper. As we think over the next 5-10 years, a lot of the competition that they will have is facing some pretty skilled companies. As you think about them fighting against the Broadcoms, Marvells and the broader connectivity players of the world, what do you attribute their secret sauce to? What do you think they're going to be able to tout the most in terms of trying to compete in what's likely a more intense competitive environment over the next couple of years here?

Specialist:
Sanjay, Jitendra, and Casey, the three founders, have been working together for an extremely long time at TI. They were the ones who founded the FTP, I think it's FTP business, which was TI's very successful automated Ethernet chip, and that whole business unit. They essentially had already started a very large company under TI. I know from whatever enough, like start-up stuff, the team is critical. The team isn't cohesive, it falls apart. These guys are like in lockstep with each other. They each have their strengths and each one of them focuses on those strengths. They complement each other very, very well. The reason why they left TI. At least I was very close with actually Sanjay.

Analyst:
I was along for the entire IPO journey, but I definitely want to hear your thoughts because I get their practiced narrative and I'd love to hear about under the covers and the greatest of the former TI dynamic. I know what you're talking about in terms of these guys.

Specialist:
Yes. The story I was told was they were going to do this under TI, but TI was such a big and slow-moving company. I think it's mostly Sanjay with Jitendra. Casey is just the pack horse, like he is just a genius when it comes specifically to firmware. All the development of all their software like the Cosmos and all that stuff and all their firmware development integrated with their chip and making their chips so flexible, that's 100% Casey. The visionary, I feel like is a combination, I'd say it's largely Sanjay and then Jitendra has his strengths. Sanjay is the visionary, I feel personally in that whole group. He's a genius when it comes to marketing and strategy and all that stuff, like he just is. He saw this opportunity coming up years ago, 2017 and away they went, and they knew they could be much faster than it's staying under TI and obviously, a lot more lucrative if not staying under TI.

That's how that all happened. Then as far as like continued execution, they're just execution machines. Those guys are like wonderful. They know what they're doing and they're powering forward. They do things a bit differently than I've noticed compared to a Microchip or a larger company, they're extremely customer-focused, like extremely, whereas we all know Hock. Hock is very much like you buy what I make. That's sort of like Hock's attitude. These guys are much more, let's become family, let's become an extension of your engineering team. They would actually, not so much the family side, but they verbalise like let us be the extension of your engineering team and let us help carry your load. Let me lift you. They, in my opinion, provided way more like free services, I would say, like having their engineering team with their FAEs could go through the schematics. Like very, helpful help that I don't think a lot of the other guys do. They're obviously more nimble and all that kind of say, they can make quicker decisions. They are not afraid to spend. Of all the companies that I really knew of when the semiconductor shortage happened, they never had one. Now granted their volumes weren't as high and stuff like that, but they're not fearful to spend.

Actually, I just finished listening to their earnings call yesterday, their inventory went up. There was questions about their inventory. That's because, they said it in the call, they want to be able to service their customer better than anybody else. They are so customer-focused. They're, let's say, the Burger King of the semiconductor industry. They make it your way. I think they're going to continue to be successful because of that mantra. It's not only just a business relationship. It's really like it is, you feel almost like family with them. I think they'll continue to execute just because of who they are and their vision, and they're going to continue to have great relationships because they understand the value of their customer, and they don't want to [upset] them by raising the rates on their ASICs because they won't buy something else. Not that they do ASICs.

For the last two decades, people were like, I hate Broadcom, I want them out because they have so much pricing power. The company that I would say I'm more concerned with is really Marvell. They seem to be much easier to do business with. I know some of the people over there. They're all good people. I know quite a few people over there through all the acquisitions and stuff over the years. We've got good people and everybody seems pretty happy over there. Marvell, I would say, is a bigger competitor than Broadcom would be.

Analyst:
You heard the earnings call, they talked about custom ASIC [application-specific integrated circuit] growth will be bigger than Nvidia-related growth in the first half. To me, I think that that's a function of two things. One is, that customers are waiting to buy Blackwell and there's a little bit of pushout. There's a bit of an air pocket, but then that's compounded by at Hopper, Astera got designed right into the board, and so they could ship with Nvidia shipping to the ODMs [original design manufacturers], whereas with Blackwell, a lot of the system integration is happening at the ODM level. The sale motion is shifting from, maybe not directly to Nvidia, but in large part, like the big focus for the Hopper cycle was getting designed in and getting on the board. From there, it's like you take your foot off the pedal and you ride the wave, whereas now it's, you have to talk to the hypers and ODMs and ensure that the retimers are still present there.

What are the frictions that you'd expect from that shift in the sales motion? Should I be concerned that there are some challenges as it relates to having as big of a presence without being right on the reference board and thinking about the fact that they need to be omnipresent across all the different hypers at the end? How are you expecting that transition to impact some of their near-term Nvidia demand because now they have to sell at a different point than they had in the last gen 5 cycle?

Specialist:
I would say they still had to sell similarly in the gen 5 cycle because from the boards I personally know, all the stuff I worked on was directly with the hyperscalers. There was like large volume from that, particularly on the retimer, which is no surprise. They're saying most of their revenue is from retimer stuff. I feel like the whole Nvidia entrance into modular board-level stuff was really just icing on the cake because they have deep relationships with all the hyperscalers as well as all the chip guys. They co-

wrote the CXL spec with Intel. Intel is an investor. TSMC is an investor, so they get, well, I wouldn't say preferential, but they definitely get a seat at the table for TSMC because like it's all interconnected. Jensen went on their IPO, whatever you call roadshow. They're very well-connected in Nvidia. The investors of Astera were also the investors at Annapurna, which was bought by AWS. There's a tight relationship there. There's a tight relationship with DigiKey. I probably won't go into, but they have a very tight relationship with DigiKey, which I don't know is that helpful necessarily with the hyperscalers.

The more, putting air quotes, family connections you have, it's like the less friction it is to do business. I don't really see much, if any, shift away. Certainly, Nvidia, I think their volumes will go down for the retimer, which obviously means that Astera will for that part of the business. They're still like locked and loaded at all of the CSPs, like I don't think it's going to impact them that much, that particular scenario. I will tell you what I know is going to impact them is they were first to market at gen 5. During the semiconductor shortage of whatever it was, 2020 or 2021, the hyperscalers learned they cannot be sole-sourced on anything. It wasn't just the hyperscalers, it's around the board, like Teslas couldn't be shipped because of a resistor or a capacitor. Server racks and systems couldn't be deployed because of some minute little USD 0.01-0.05 piece. I saw that or I heard about that stuff with Microchip because Microchip supplies a lot of those like what I call popcorn, for the board.

I use Tesla as the case because it's just easy and more visually easy even though, whatever. You're not going to be able to ship a USD 50,000-100,000 Tesla because you can't get like a USD 0.05 oscillator. That's a scenario before I went to Astera, like I was doing timing stuff. Tesla was paying 10x, 20x, 30x for a USD 0.05 piece just so they could ship more Teslas. All the products that were based on semiconductors really like struggled during that whole process. Especially, the CSP guys, they're like never again. Astera cleaned up the market at gen 5, like on all the first projects that were being released, which are typically the large volume projects. They're doing it again on gen 6. They will be first to market. Their strategy is first to market. They will capture a lot of stuff on that. Now the CSPs having learned from that experience, they are now adding second sources. Now here comes the Broadcoms and the Marvells.

The things that Broadcom and Marvell, the leverage they have over Astera is their ASIC divisions. The only one that does ASICs really are Broadcom and Marvell. I think Marvell bought IBM and Broadcom bought like LSI, I think it was or something. Anyway, those are the only two guys that can really do ASICs anymore. They codevelop a lot of this stuff. I think there was a press release that AWS and Marvell just signed some deal or something like that. Those are going to be their biggest competitors, Broadcom and Marvell. It's going to be between Hock and like I don't think then they're not going to sign up Broadcom and Marvell. They're going to sign up one or the other typically. I don't think they're going to give their secrets out to both companies. They're probably going to have one ASIC guy and then as a second source, Astera.

Now how that all shakes out with share at each individual CSP, it all depends. Again, since Astera is first to market, obviously, they're going to see the lion's share of the high-volume stuff. Then as the other guys come up and release their products, they'll get released on... When they get to redesign the board six months or a year later, like they'll get that. Then maybe after that that's when they'll see more shift in market share towards the ASIC guys or they keep it at 50-50 or whatever. Astera knowing that they are somewhat at a disadvantage with the ASIC guys, that's why you see the high inventory levels. If there's any upside, they want to be able to capture it because they know they're at a disadvantage. At least this is how I see it. That's why they're so customer-friendly because they have to, in some way. They don't have the clout that those two guys really have.

Then specifically on the switch, that was a bold move. The ASPs on the switch are a lot higher than the other, I wouldn't call them popcorn chips, the other stuff that they sell, but they're like more commodity chips.

Analyst:
250 vs 50.

Specialist:
Exactly. It's been very interesting to see the switch thing. They were developing that when I was there, and it's very different than what I was hearing what it was going to be. I'm glad that they changed it because the thing that they were going to originally build was I was like no one is going to want that. It's good to see that they added fabric to it.

Analyst:
On the share dynamic for retimer, I think everyone is trying to debate what the second source is going to look like. Speaking of Astera Labs, they think they can keep 70-80% share. Over the longer term, through the next five years, I don't think anyone believes them on that. Can they keep the majority share? Can they have a 50-60% share of the retimer market when they have Broadcom and Marvell competing? Do you think that primary share is maybe something closer to 40-50% and then the rest of it will get cut a few times across Broadcom and Marvell and to a lesser extent, Microchip, Credo, Kandou or other players? How are you thinking about what primary share can look like for them over a five-year view?

Specialist:
Let's just take gen 6 because when you throw in gen 7, that's like a whole another, I'm just going to say gen 6. They're going to get majority share at the beginning. It depends on when the other guys release their chip. They're going to get majority share because they're first to market. Then over time, as new things develop and like its annual contracts, I would say it's going to start eating up share maybe one year out. As you alluded to, there's so many different things that can impact that. I think they're doing everything they possibly can like with the large inventories and all that stuff to be as flexible as possible for the customers. To be honest, you're probably aware of this, but the customer forecasts are crazy. You might as well not even get one. You probably have a better chance guessing what you actually get from the customer. I think in the first year, maybe 1.5 years, they'll be majority share, and that could look 70%, 80%, 90% depending on the gap between the time they release vs a Broadcom or Marvell.

Then over time, it's going to be like a price and leverage play. Obviously, the other guys have more leverage because of the ASICs. It's interesting that they're bundling things, like they're now bundling things and they launched DesignCon where they said the performance is better when you use all Astera products, which isn't surprising. They are going to look at ways and have to look at ways to prevent or mitigate the risk of the lack of leverage that they have because of the ASICs. I don't think it will go much below 50-50, I guess what I'm saying, I think for the first one-and-a-half years, it will probably be 80% or more. Then the follow-on years, 50% is usually what happens. Now that they're adding the switch chip and that's a much stickier thing. They're going to have more leverage than before they had the switch.

Analyst:
On the switch, Trainium2 for Amazon has already started ramping. They missed it. Trainium3, maybe they get some opportunity there. For the most part, the P-Series switch that they're trying to sell, it's going to be Nvidia-based. Within the Nvidia opportunity, it's not any of the reference design stuff. Anything that Microsoft buys is like the standard spec. You get the whole thing spun up for them. They don't want to do any of the actual system integration. Microsoft feels like an unlikely opportunity. You got the other guys, mostly Amazon and Google, maybe to a lesser extent, Meta, who are going to do some customised systems where they don't use the great CPU and that's where the opportunity exists.

You have a USD 1bn market. Like you said, Broadcom's got 95% of it. Astera can't compete for any of the reference divine stuff, which is probably a pretty good chunk of it. How are you thinking about how much share they can take over a 1-2-year timeframe, especially considering that I'm sure Broadcom will do some price gouging on their chip because they're willing to take the hit to tell Astera to kick a rock? How are you thinking about what share aspirations should be on a reasonable basis for Scorpio as they look to ramp it up over these next 24 months?

Specialist:
To be honest, I don't think this version of Scorpio is going to be like a wild success. I think the next version of Scorpio is when they're going to really start taking market share because they had to get their feet wet. Doing fabric switches are no joke. I don't know if that's the company they acquired or whatever that they mentioned in the call, but doing fabric switches is no joke. Then also just what I know and what I've seen at least at Google for fabric switches, like that wasn't enough port counts for their high-volume designs. They wanted much larger port counts. However, they were asking about smaller port counts. It's a little unclear to me, are there going to be new topographies released to utilise these smaller port counts? I don't know. I don't know enough about that yet. I know that in the past that like Meta, their boxes look very different than what Google's boxes look like, which look very different than what AWS does. I'm hearing shifts in the topography of stuff and maybe that opens up the opportunity for a 64-port switch. My gut is telling me it's more like they just want to be able to do it. I don't know how much I can say, the original reason they designed the switch was for a specific DSP.

Analyst:
AWS [Amazon Web Services], yes.

Specialist:
Okay. I didn't know, what I'm supposed to say. Where they're finding like other use cases, I'm not so sure because I have not personally seen use cases. I've seen people kicking the tyres for a slow port switch. I don't know if it's some of these other players like, I don't know, OpenAI's thing that they're doing or whatever Elon is doing or some of these other guys, Cerebras. I don't know if they're doing that out. I would say their market share is going to be pretty small. Just in general, everybody's first revision or first entry into a new chip typically doesn't go super smoothly. There's many instances. Even the Pixel. They got rid of Pixel or Google got rid of Qualcomm and went with their own chipset and got janky for a while.

I think this is more of like send a message to the market, proving that they can be more than just these commodity chip type things and that they can build bigger chipsets. I guess I would say not much market share, maybe 10-20% market share, I would say, not revision, but like the first chip, Scorpio 1 (ph). I think they'll be able to take more and more of that on at Scorpio 2 and Scorpio 3. I think it's going to be more delicious later on. Then as far as Microchip goes, like I don't see them coming back. It's unclear to me if Marvell, I don't keep up with Marvell as much as I probably should, but Marvell is doing the business.

Analyst:
They did PCIe 7 [sic] SerDes demo DesignCon and the way they configured it, it looked like it might be a switch chip. I feel like they're going to miss gen 6, but maybe we'll hear about them at gen 7 a little bit more.

Specialist:
Yes. The last time I talked to those guys, they were passing on the switch chip, probably because it was Microchip and Broadcom incumbency. Then right now, that Microchip has just dropped out. There's a need for a second source.

Analyst:
On the port count dynamic, Astera's pitch is like if you're a cloud guy such as Google, Amazon, Microsoft or Meta buying these 144-lane products and then you think about where you're actually interconnecting on the head node, whether it be a couple of NICs [network interface cards], SSDs, CPUs, none of that actually adds up to 144 lanes and maybe you need 100-ish port lanes at the end.

As you think about a lot of the customer set utilising those 144 lanes in the past, would you agree that a lot of them are underutilising their port counts at the end and so that value proposition makes sense where Broadcom is stuffing 144-lane down their throats because they got 95% market share, so they don't care if you're underutilised? It sounds like some people are kicking around the idea of these slower port count switches, but at the same time, you mentioned Google would need more lanes than just 64. I'm trying to find a balance between whether or not underutilised lane counts are a real thing or if the hyperscalers do need more than even 128 lanes if you just bought two 64-lane switch chips. How I should be thinking about that?

Specialist:
Again, I haven't worked with Google in six months or so, but I will say that they asked for a switch chip with much, much higher port counts, like absurdly high port counts. I didn't get into the details of what it was for or whatever because they're very secretive about what they're designing, obviously. Then there was some start-up that claimed to have a very high port switch chip, but I'm not sure where that ever went. 144 sounds high. What was that?

Analyst:
Maybe Enfabrica.

Specialist:
That sounds familiar. I don't remember. Enfabrica sounds familiar. Again, there's no way that Google is going to put all their eggs in a small start-up's basket. 144 even sounds high based on what I've seen. We discussed, like Broadcom if they make something too big. They're just like, "You're going to buy what we have." They can because there's no one else to go to.

Analyst:
Microsemi fell off, so they could just sell 144. Now if 164-lane or 264-lane, so total 128 is good enough. I don't know, Broadcom is probably selling those 144s for USD 600 and Astera selling 64 for, I don't know, USD 200-250, so then two of them are still only USD 400-500 vs the USD 650 that Broadcom is selling for. Is that a good enough value proposition? We'll see how the product works.

Specialist:
The mantras I've seen is like the less chips, the better. They're just having such a hard time creating these boards. Having said that, there's a lot of liquid-cooled stuff coming out, which makes a dramatic difference in signal integrity and power improvements. I know there's a lot of liquid cooling starting to come on the market. That's another thing that needs to be considered.

Analyst:
Retimer's main thing is it's solving for signal integrity. As you think about PAM4 [pulse amplitude modulation level 4] optics, linear equalisers, liquid cooling, pluggable optics and co-packaged optics, everything screams like retimer will peak out at some point. How do you think about the timing of that where the amount of retimers on a given board falls off precipitously because signal integrity is solved in different ways? How are you thinking about it, is it next year, the year after or is it longer than anyone thinks?

Specialist:
It would (talking over each other) the year.

Analyst:
As I think about a few years out, can you explain that as all the naysayers that are so juiced up on optics think that it's going to be way earlier? Can you talk me through from your perspective why that's not likely to be the case, at least as quickly as maybe some of the most pessimistic people towards retimed solutions might be?

Specialist:
There's multiple factors. I know a lot about fibre optics. Gen 6 is here. Things are being actively developed. Gen 5... Forget it, that's done. Gen 6, unless there's a whole bunch of optics coming out very imminently or there's like a massive change in architecture, I just don't see it happening that fast. I could see it at gen 7 maybe, but I just don't think so at gen 6 for, again, multiple reasons. There's this whole promise of silicon photonics and stuff. I don't think they're there yet. I think it's more there on the Ethernet side, but not on the PCIe side. UALink could come in because I'm a little unsure how much of that's going to replace. If it's going to go optical, it's going to start off copper, blah, blah, blah. The cooling definitely makes a difference for sure. Co-packaged optics, I have to look into that more. I don't know, I used to hear so much about it, like the COBO and all that stuff from Microsoft a decade ago and nothing ever happened. I don't see it happening for gen 6, certainly not at volume. Now could they have some smaller use cases, like corner use cases or low-volume use cases where that takes over? Sure. At 6, I just don't see it happening at 6. I just don't. That's on the optics side. The cooling will make a difference. I just am not sure how much.

Analyst:
In the past, Nvidia would drop a new chip, 30-50% performance improvement with that bandwidth jumps a bunch every two years. Now Rubin will be here by, I guess, by the end of this year or early next year.

Specialist:
Yes, I don't know how they're doing it.

Analyst:
In your mind, is gen 6 good enough for that as you think about the past jumps in performance and bandwidth on a chip-by-chip basis? If not, the thing I worry about is we just got gen 7 spec for PCIe a little while ago, a controller came out. It takes some time to spin up and tape out an actual controller. I wonder if Rubin is going to be ramping in late '25, early '26 and it needs something better than gen 6, should I be a little bit worried that that opens up a little bit of an air pocket in terms of being first-to-market or even being there in time for the next wave of new generation GPUs?

Specialist:
As far as the retimer goes?

Analyst:
Yes, as I think about Rubin.

Specialist:
Astera is like in lockstep with Nvidia. I would be shocked if Astera lost their primary position because that's what they need to compete. They need that to compete with Broadcom and Marvell. They need to capture that early design win at volume. I wouldn't be worried about that per se. I see what you're saying, if it stays at gen 6, then they won't necessarily have that advantage anymore. I would suspect it's

going to be gen 6. Now that I've thought about it, yes, I could see there being a little pocket there. They're not going to get 7...

Analyst:
I agree with you. Gen 7 feels unlikely and like to be out here by the end of this coming year if PCIe moves slow.

Specialist:
Slower, yes. No, it's not going to have gen 7. I would be shocked if it had gen 7. No, you're right. I could see that being a little pocket for them. Here's the other thing, though, is Nvidia is starting to take on Broadcom as well with their announcement of DSPs and coherent or whatever. Obviously, they would already be competing with Marvell. They would be competing with Marvell too with the (audio distorts) anyway, but both of them have those technologies. It's like Nvidia is moving into Broadcom and Marvell space. I think that Astera is well suited because they're not going to compete with Nvidia on these new chips.

Analyst:
The bluebird opportunity for the switch chip seems to be the alternative to an NVLink for hyperscalers in terms of back-end interconnect. How challenging do you see that being, especially considering a team that is only putting their big boy pants on as it relates to switching technology very recently? How much wood is there to chop to be an effective alternative to hyperscalers when they don't want to pay the NVLink tax, so to speak?

Specialist:
I think it's twofold. I wouldn't be surprised. My gut is that UALink is probably going to take on NVLink. It's not just that. Everybody is trying to get away from NVLink. Again, I don't think it's going to be radical adoption. It's going to be like a phased thing, like get feet wet this generation and then move forward next generation, buy market share or whatever. I think UALink is going to be competing against NVLink, or do you not think that?

Analyst:
I want to see when they come out with the spec in March. The challenge I have is that Broadcom is [dismissing] UALink [ultra accelerator link] in favour of Ultra Ethernet. Then it's just going to be a competition between the two. The good thing going for UALink is nobody wants to buy from Nvidia just as many people don't want to buy from Broadcom. Then UALink fits the value proposition of being something else. I want to see what the approach is going to be and what the spec says, and that will give me a better sense of at least how far away we might be from them being a viable competition. It sounds like we'll hear about that in March.

Specialist:
No, for sure. My gut is that or my read on this is UALink is going to take over because for what you had already mentioned, Broadcom and Nvidia. No one wants proprietary. Everybody wants open source now. That's the mantra of everybody now, open source or more open source than proprietary.

Analyst:
There we go. It's a nuance.

Specialist:
Exactly. Nvidia has had, and you already know this, but like, let's just say, virtually 100% market share, there's only one way to go down. They can't gain any more market share from where they're at. It's inevitably something else has to take over. I see UALink. Again, to your point, yes, let's see how the spec

rolls out. If you just look at the trends of the CSPs, they're moving towards their own ASICs. They're moving away from merchant silicon as much as possible. They've really homegrown all their teams for chip development. It would seem to fit with the trend for UALink.

Analyst:
I'm excited about it, but it's definitely early days. I'm intrigued to see what we'll hear more about that over the next coming months because I'm sure there's going to be a lot.

On Scorpio, I don't know if you caught the comment from the team on the earnings call, but talking about it being its largest product over the next couple of years. I feel like you can interpret that two ways, retimer revs were like, I don't know, USD 300m this year, going to grow next year and probably the year after, but at lower rates. Either way, do you think Scorpio is becoming the biggest product, that it's going to be a USD 500m-plus business? Or do you think it's more that retimers are going to fall and that Scorpio is going to ramp into something less than that USD 500m, but it will still be the biggest, but retimers are smaller by the time that it becomes the biggest?

Specialist:
I don't feel like retimers are going to fall as hard as other people may believe. I still think retimers are going to be around in gen 6, especially because of the PAM4 and the doubling of the rates. I just think it's technology that's been around forever. I think it's going to take more time to get rid of it. Do you know what I mean? I think it's going to flatten out before it's going to take a nosedive. I don't know that I necessarily believe that the switch chip is going to be that big. I obviously could be wrong, but I don't know. Unless there's something that I'm not aware of, like a massive topography change or I don't know the 64-port is just like I'm scratching my head on it.

Analyst:
It's a good news, bad news kind of dynamic. The good news is, retimer is not going to fall that much, bad news is Scorpio still probably won't get there no matter what.

Specialist:
I don't know. Who knows? I could be totally missing something that they have an exclusivity deal with so and so. They're involved with everybody. I think one thing that they should do, and someone asked the question, how are they working with all these new entrants to the AI chip market? It seems like everybody is developing a new AI chip or a GPU chip or whatever. I know they're doing well with all the incumbents, but there just seems to be so many new players backed by big money. People are going to pour money into that, Sam Altman, Elon. I'm sure there's others. I think Bezos has got his own AI chip. Are they really addressing that or are they all going to win because they're all going to decide that they all need exactly what they're developing and they would never pay the other one for their IP? There's definitely the billionaires don't want to be outshined by other billionaires' scenarios. I just don't know if they've been fostering those relationships given how much they have already with the main guys.

When I was there, they weren't really addressing anything that was ARM related. I know there's been a big shift towards ARM over x86 in the data centre. Now I don't think that's for AI, but like are they interacting with Ampere or some other ARM base? Then you throw in RISC-V. I know RISC-V is further away, but are they addressing those markets, too? Then if you really want to blow your mind, what are you going to do about the automotive market? We all know that our car has so many data centres. I wouldn't be surprised if they start going after that market, too.

Analyst:
Especially when BYD made a big splash in the market talking about their intelligent driving just this week.

Specialist:
Wait, who did that?

Analyst:
BYD.

Specialist:
That's what I'm saying, like ADAS, I don't know what we're at now, ADAS 3 or whatever. That's going to be another huge wave.

Analyst:
Four product families – PCI retimers, Ethernet retimer paddle cards, CXL [Compute Express Link] controller chips and fabric switch chips, where do you think they, as you think about adjacencies would fit best, that if you heard like, six months from now, they announced XYZ product that you'd be hyped up about it because it is a good adjacency with a strong market, kind of secular dynamic and that they'd be able to grow nicely and layer on to the four families they already have"

Specialist:
They started doing it with optics, but I'm biased because that's my wheelhouse. They announced the PCIe optical cables. Margin-wise [its bad]. As far as a like a quick increase in revenue, that's a no-brainer. What else? As far as markets go, automotive, I don't see why wouldn't they just port a lot of this stuff over to automotive somehow. I'm trying to think what else. I know they have aspirations to a bunch more stuff also with Richard Ward as being their standards guy like it's Richard and Chris Peterson, formerly from Meta. Richard is an optical standards guy. They obviously have some sort of aspiration for optics. Again, silicon photonics is getting pretty hot now. I see a lot of growth in that for multiple big chip companies. Something related to optics, I think that they'll continue pushing down that path. It will be more on a partnership side. It won't be on like, I'm not going to get all the CAPEX and all that. The CAPEX for optics is like insane. Again, lower margin stuff probably, but room to grow.

Analyst:
I definitely think they're looking at it.

Specialist:
It's the classic interconnect. It fits well with their core competency. There's a lot of consolidation happening now in optics, so anyway.

Analyst:
If you're Sanjay, Jitendra and you're unable to sleep because of something about the future, what do you think is the most concerning unknown for their longer-term trajectory?

Specialist:
Short-to-midterm is just like what I've said, the ASIC stuff, like not having that leverage that the other two do. Longer term, I think it's like how long are these interconnect... If they're able to do all this stuff on a board through whether it's like cooling or whatever, where are these interconnects going to go? Are they even going to be needing for these interim interconnects like the retimers or whatever? Like I said, I can see the retimers maybe going away in the future or at least not needed as many as much, or copper is not good enough, all this COBO and stuff. I just feel like in some use cases, we're getting to the end of economic viable use of copper. Again, I'm biased towards optics because it's what I studied in engineering school and that's what I did for the first half of my career with optics, I don't know. I just see now that you've got Ayar Labs that are doing the optical I/O's flagship.

I even worked on an optical FPGA back at Altera, whatever, Intel 15 years ago. This thing has been around for a long time. The idea of this concept has been around for a long time. I think it's finally getting close to being more adopted. I think they're going to be at a loss if they don't get into an optical space, more so than they have already, which is like the PCIe cables. I'm going to say silicon photonics. That's my answer for a lot of this stuff, photonic chips.

Analyst:
We'll see when it comes.

Specialist:
Yes, I know of a lot of companies that are working on it.

Analyst:
Tower Semi has already got USD 100m in silicon photonics wafer foundry revenues. That's a canary in the coal mine for future products.

It's exciting and nerve-racking at the same time, just as I think about Astera because I hope they throw the needle on it.

Specialist:
In fact, I'm too impressed. I'm trying to not get involved, but I agree.

Analyst:
We'll see. I think we're close. It's all about figuring out exactly when it will happen, but it's worth thinking about the (talking over each other).

Specialist:
It's happening. It's definitely starting to happen, for sure.

Transcription ends